

Preliminary evaluation of a Tag-based Knowledge Condensation Tool in Agile and Distributed Teams

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Abstract—An important challenge in Agile Global Software Development (AGSD) is architectural knowledge vaporization, i.e., the loss of technical knowledge due to a lack of documentation. In a previous work we identified that technical knowledge is usually available in unstructured textual electronic means (e.g. chat, mail, blogs, etc. – also known as UTEMs) used by AGSD workers during development, although not necessarily easily accessible. Also, we proposed the concept of Knowledge Condensation as a means to recover knowledge from UTEMs, by introducing a structure to their contents along with means to retrieve knowledge. In this work we present our first step towards a Knowledge Condensation tool, through a prototype tagging mechanism to structure architectural knowledge during UTEM interactions. We evaluated the prototype and obtained the following results: (1) a high perception of usefulness and a high projected intention of use; and (2) the identification of 6 tagging behavior profiles from participants. These tagging behaviors can be grouped as (i) participants who tag as expected, (ii) participants who tag more than expected, and (iii) participants who tag less than expected. These results suggest that it is important to cope with “little or no tagging” and “excessive tagging” behaviors to implement knowledge condensation based on UTEM tagging. Furthermore, we identified that tagging could be a good option to structure UTEM interactions, and to retrieve architectural knowledge from UTEM logs. Future work includes improving the tagging mechanism prototype based on the participants’ suggestions and overall evaluation results, and implementing a searcher prototype to further evaluate the concept of knowledge condensation based on tagging of UTEM interactions.

Keywords- agile global software development; architectural knowledge; unstructured and textual electronic media; knowledge vaporization, knowledge condensing.

I. INTRODUCTION

Nowadays, the number of agile and distributed (or global) teams are increasing day by day. In these kind of

teams, Agile Software Development (ASD) and Global Software Development (GSD) clash due to their distinct nature and distinct principles. On the one hand, ASD was designed for co-located teams, while teams are distributed in GSD. Also, in ASD knowledge sharing is preferably done through individuals interactions to maintain a rapid development [1], while in GSD is preferred to share knowledge by structured means (e.g. documents, wikis, knowledge repositories, etc.) to decrease the effect of the four inherent distance in this paradigm (physical, temporal, cultural and linguistic) [2]. These differences lead to problems in knowledge management [3], [4].

A highly valuable knowledge in any software development paradigm is related to software architecture; from the most granular (detailed design, technical knowledge) to the most abstract (architectural patterns, good architectural practices, etc.). This kind of knowledge is called Architectural Knowledge (AK), which also includes design decisions and design rationale [5].

In agile and global teams (AGSD teams) AK management (AKM) [6] is a challenge, due to the preference of personal interaction over documentation, and to time reduction to write proper documentation, which negatively affect the coding phases [7]. So, in AGSD tacit knowledge tends to predominate over explicit knowledge [4], [8].

AGSD conditions lead to documentation debt, which refers to the lack of documents in the software development cycle [9]. AKM is affected by documentation debt when technical documents are missed, e.g. architecture design, data models, architectural rules, etc. We can say that the main reasons of documentation debt in AGSD are: (1) the agile principles which suggest to make “lighter” and “sufficient” documents, but the pressure to complete a sprint causes documentation omissions [10]; (2) agile developers are reluctant about documentation, since they consider it as a secondary and non-creative activity [11].

A consequence of documentation debt in any agile method is AK vaporization, i.e., losing AK caused by

improper technical documents [12]. Once AK has been vaporized, the popular way to recover it is through source code analysis, which is time consuming. Further, the knowledge obtained is generally unstructured, incomplete, and inconsistent [13], which does not guarantee that the software evolve as it was planned at designing time.

Analogously to the AK vaporization concept, in a previous work we proposed [14] the concept of AK condensation. This concept is based on the AGSD developers' preference of using text media to interact with remote members, since these means (which we will call unstructured textual electronic media – UTEM) reduce communication problems caused by language differences [15]. In another previous study we found that UTEM logs contain important AK for AGSD developers [16], but this knowledge is unstructured, inaccessible, dispersed and prone to be vaporized. To avoid knowledge vaporization, UTEM interactions have to be stored in a common and accessible repository, and there should be a classification mechanism to structure the UTEM interactions; so that AGSD team members could find AK aided with this classification scheme.

In this paper, we present a classification mechanism based on manual tagging (as in social networks) performed by developers during their UTEM interactions. This way, these interactions will be tagged and stored in a repository, so AGSD team members could search for AK based on the classification tags inserted. We also present an evaluation of the UTEM interactions manual tagging, where usability, intention of use, and tagging behavior were obtained. We discuss the results and finally we present our conclusions.

II. TAGGING MECHANISM

Tagging has been used in Integrated Development Environments to find and remember information on source code, with good adoption by developers [17]–[20]. Also, auto-tagging mechanisms have been developed to relate emails with requirements, but only with a 70% of accuracy [21], which could be unacceptable in a production environment. Thus, we think that AGSD developers could also adopt UTEM tagging.

However, one of the characteristics of tagging in social media is the excessive creation of tags, known as tag explosion [22]. For our purposes, this is a problem since the author of the tag could be the only one who knows its exact meaning, i.e. interpretation difference and locality of tags

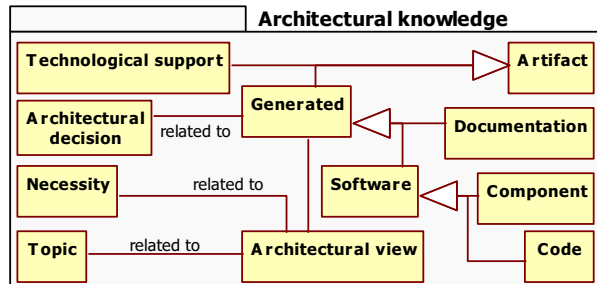


Figure 1. Ontology that represents the involved aspects in AK articulation through UTEM in AGSD teams (reproduced from [14])

could be present [22]. Thus, knowledge retrieval could be problematic if it is based on a classification scheme produced by a mechanism without tag creation control.

Thus, we developed a prototype mechanism to help AGSD developers to tag UTEM interactions. This mechanism uses a list of tags that are defined a priori by the group of AGSD developers (users). All tags must be related to a set of meta-tags that are embedded in the tagging mechanism and that cannot be edited or deleted. The set of meta-tags is based on the results of a previous empirical study, where we analyzed a set of UTEM interactions among AGSD developers [16]. These meta-tags represent the AK that AGSD developers share, according to our study (see Fig. 1). The aim of relating custom tags and meta-tags is to provide a more abstract and structured way to store and find AK. For example, the #pythonTricks tag could be related to the #Code meta-tag, so that when a developer wants to recover AK but he does not remember the exact name of the tag, he could use the meta-tag to conduct a first search.

The tagging mechanism was developed in Javascript and it was added to a custom instant messenger (IM) tool (developed ex professo). The tagging mechanism is triggered when the hash character and a letter are typed on a UTEM interface (e.g. “#a”). Then the mechanism shows a list of the proposed completions based on the preloaded list of tags and meta-tags. Then the developer selects the tag that better fits to the interaction topic, and the tag is inserted on the text of the current message (see Fig. 2). Either the mouse or the keyboard can be used for tag selection (using the up/down arrows to navigate the tag list and tab to select it). When an inexistent tag is entered, i.e. a tag that is not part of the preloaded list of tags and meta-tags, the mechanism deletes that tag, as for the current implementation new tags cannot be added to the custom list through this mechanism. New tags are added using an administration module during Sprint planning or Sprint retrospective meetings.

III. EVALUATION DESIGN

We conducted an evaluation to obtain the usability perception and the intention of use of the tagging mechanism. We also observed the tagging behavior of participants during their interactions. Participants were 30 developers from five Mexican software development companies (avg. age = 27.2, std. = 3.2) with 4.7 average years of experience on software engineering (std. = 3.8), 2.3 years of experience on ASD (std. = 1.1), and 1.3 average years of experience on Distributed Software Development (std. = 0.8).

The evaluation was designed as follows. Firstly, we explained to each pair of participants the problem of AK

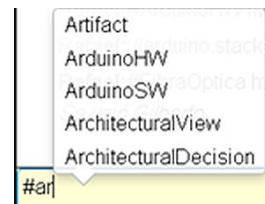


Figure 2. Tagging mechanism (autocomplete example)

vaporization in AGSD, and the proposed solution based on UTEM interactions tagging. Then, we presented the custom IM tool and the proposed tagging mechanism to them. They freely used it during two minutes to realize how it worked.

Later, we asked to the participants to assume to be in a scenario in which both of them work for different software companies (A and B), and that they are collaborating in the same project. Company B depends of a RESTful service that is currently being developed by company A, and there is no actual documentation on that service yet. The developer from company B has to obtain the AK of the service from the developer from company A. With that in mind, we asked them to use the custom IM tool following a chatting script (one per participant) in separated rooms. According to the chatting scripts each participant has to ask three questions related to the RESTful service. Also, the scripts had three parts where we suggested them to use a tag, without specifying the exact tag name. With this aim, we preloaded four custom tags to the tagging mechanism, which were related to the scripts topic. These tags were: #Security, #ErrorHandling, #UserStory, and #BusinessRule. It is worth noting that we said to participants that they were free to use the tags, if they thought that any tag fits in the conversation. The complete chatting activity had an average duration of 20 minutes.

Once the scripts were entirely followed, participants answered SUS and TAM questionnaires. Finally, we asked participants to provide a final comment or suggestion about the tagging mechanism.

IV. EVALUATION RESULTS

A. Tagging behavior.

We counted the number of used tags per participant, distinguishing between custom tags and meta-tags, and identifying when they were semantically correctly used. Based on their behavior, we identified six different participant behavior profiles:

1) *Expected*. Participants who used only the three tags that were suggested, i.e., each correct tag, one instance per tag.

2) *Expected + repeated*. Participants who used the three suggested tags, and that repeated its use in another place in

TABLE I. SUMMARY OF NUMBER OF PARTICIPANTS PER PROFILE.

Participant profile	Number of participants	
	Correct	Incorrect
Expected	7	0
Expected + repeated	10	1
Expected + meta-tags	7	4
Less than expected	6	0
Less than expected + repeated	2	1
Less than expected + meta-tags	3	1

the conversation, i.e., each correct tag, more than one instance per tag.

3) *Expected + meta-tags*. Participants who used the three tags as suggested, and that used meta-tags in other places in the conversation (each correct tag, plus meta-tags, one or more instances per tag).

4) *Less than expected*. Participants who used less of the three expected tags.

5) *Less than expected + repeated*. Participants who used less of the three expected tags, and that repeated its use in other places in the conversation, i.e., more than one instance of any used tag.

6) *Less than expected + meta-tags*. Participants who used less of the three expected tags, and that used meta-tags in other places in the conversation (one or more instances per tag).

We observed that only 7 participants were classified as having the expected profile (see Table 1). Also, we observed that 17 participants used correctly more than the expected tag instances ((Expected + repeated) + (Expected + meta-tags)), and only 5 used them incorrectly. Further, 6 participants used less than the expected tag instances, where 2 of them repeated an expected instance and 3 used one meta-tag at least.

We represented the tagging behavior in a bubble graph (see Fig. 3), where we observed that most of the participants used up to 5 tag instances. Also we observed an outlier case of 8 used tag instances. In the case of the tag instances used incorrectly, most of the cases are of only one instance.

B. Usability and intention of use.

In terms of usability, we obtained an averaged SUS

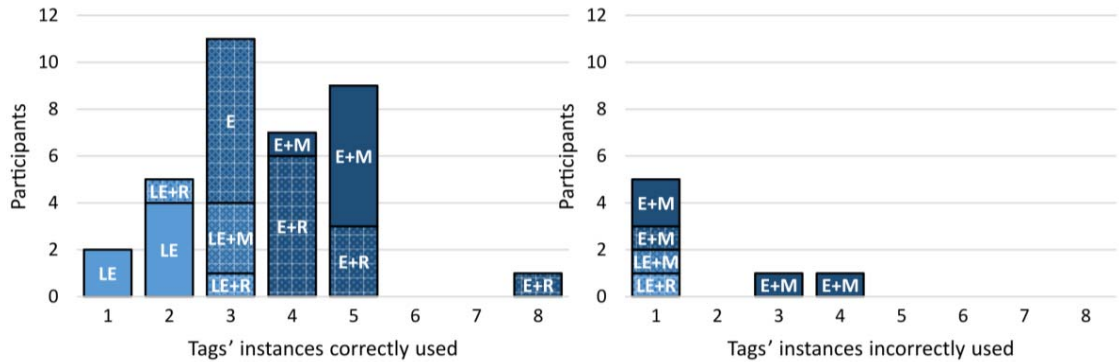


Figure 3. Tags' instances used per participants' profile (LE = Less than expected, LE+R = Less than expected + Repeated, LE+M = Less than expected + Meta-tags, E = Expected, E+R = Expected + Repeated, E+M = Expected + Meta-tags).

score of 86.66 (std = 10.98), suggesting a high usability perception on the tagging mechanism. However, Fig. 4 presents this result using a curved grading scale [23], where we observe that around 10% of the participants perceived the tagging mechanism as having a low usability (C and D grades on Fig. 4).

Concerning the TAM questionnaire results, we obtained a perceived usefulness of 6.05 using a Likert-7 scale (std = 0.89), i.e., the participants consider that the tagging mechanism is highly useful. Regarding the perceived ease of use, we obtained 6.45 in Likert-7 scale (std = 0.86). This TAM result coincides with the SUS result, suggesting that participants perceived the tagging mechanism as highly usable.

V. DISCUSSION

One of the main findings was that the number of participants who used more than the expected tag instances was higher than the participants who used only the expected tags and less than expected. This suggests that software engineers did not have any problems on tagging their UTEM interactions using the proposed mechanism. The tagging behavior of the participants could be reflecting the popularity of tagging in social systems [24], however we must explore better this possibility. Further, we observed that they really assumed their roles in the evaluation, so they were interested in tagging correctly to facilitate future knowledge retrieval.

Despite the good usability perception, supported by the results of the SUS and TAM questionnaires, participants suggested a list of possible improvements, including: (1) a customizable way to select and navigate the completion suggestion list (i.e. some participants had problems using the TAB key to select a tag); (2) a context aware manner to suggest tags, i.e., based on the interaction topic; (3) include an additional way to see the available tags, i.e. in addition to autocompletion (e.g. a tag searcher, or a tree of available tags); (4) a mechanism to identify frequent concepts and then convert them into tags. These suggestions may provide possible explanations about the low usability perception on the proposed tool by 10% of the participants as identified using the SUS questionnaire. This in turn suggests that it is necessary to keep working on improving the usability of the tagging mechanism.

Also, it is necessary to express a word of caution regarding an excessive use of tags during the interactions, as in this evaluation we had a participant who used 8 tag instances instead of the 3 required. Considering a longer chatting period (e.g. a whole day), excessive tagging could

affect the effectiveness of AK retrieval. We proposed to use tags to find knowledge, so if there are many similar tags in UTEM interactions, the search hit list could grow big, resulting in information overloading. In fact, after concluding the evaluation, about 50% of the participants asked for the search tool characteristics, and requested to consider the effectiveness of the search results, so that obtaining the desired knowledge in less time could be possible.

Another finding was that participants used meta-tags in a correct manner; despite we never explained them their meaning. This suggests that the names proposed for these meta-tags are expressive, and that they relate to everyday situations in DSD and/or ASD. However, there are some meta-tags that were consistently used incorrectly (e.g. #TechnologicalSupport, #ArchitecturalView), and others that were never used (e.g. #Topic, #Generated). In the case of the tags that were used incorrectly, it is necessary to identify a more adequate name for them. Considering the meta-tags that were not used, we consider that the evaluation scenario was not suitable for the use of those tags. Thus, it is necessary to keep evaluating the tag set in other scenarios.

Also we found cases where participants used less tags than those expected. During the evaluation, we observed that those participants had difficulties to select a tag, caused either by the selection mechanism or because they did not find any tag that matched the script's suggestion; as a result they preferred to send the message without a tag. This behavior showed us that a customizable way to select tags is important and useful to decrease the “little or no tagging” cases, as well as, a mechanism to add new tags during interactions (thus, in-Sprints). However, we must be careful with the provision of a mechanism to add new tags freely, because it could lead into tags explosion.

Further, participants from this profile seemed in a hurry to conclude the evaluation, as they did not want to spend too much time trying to perform the activity in a correct manner. In real settings, ASD developers deal with similar in-a-hurry situations, even though this kind of situations cause technical debt [10]. For this reason, it is necessary to improve the mechanism usability to correctly cope with this kind of situations.

Finally, the behavior of participants in the “Less than expected” profile, in terms of AK retrieving, could negatively affect knowledge acquisition, as there would be few (or inexistent) points to refer to when trying to find the required knowledge on the UTEM logs.

VI. CONCLUSION

In this paper we presented a prototype mechanism to aid AGSD developers to tag UTEM interactions in order to later retrieve AK through them. This mechanism was evaluated by 30 software engineers in a simulated AGSD scenario. They perceived the mechanism as highly usable and they showed a high intention of use. However, we received suggestions to improve the usability of the tagging mechanism.

We also used the evaluation to observe the participants' tagging behavior. We identified 6 participant profiles that can be grouped on (1) participants who tag as expected, (2)

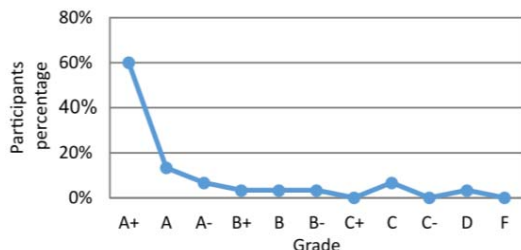


Figure 4. Curved grading scale of the obtained SUS score

participants who tag more than expected, and (3) participants who tag less than expected. We noted that the number of participants who tag more than expected was higher than the others groups, which coincides with the high perception of usability and intention of use of participants on the tagging mechanism. These results suggest that tagging could be a good option to structure UTEM interactions, in the aim to retrieve AK from UTEM logs (a naturally unstructured source of knowledge).

Also, we discussed about the importance to cope with “little or no tagging” and “excessive tagging” behaviors, as well as, the possibility to add new tags “on the fly”, in the aim to implement AK condensation based on tagging.

As future work, we will consider the participants’ suggestions and the overall results of the presented evaluation, to improve the proposed tagging mechanism prototype. Also, we will refine the meta-tags (see Fig. 1) through an additional evaluation using the improved prototype. Finally, we will implement a tag searcher prototype in order to be closer to the implementation of AK condensation based on tagging of UTEM interactions.

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